

Applying Roles

The developer's role specifically contributed to the Scrum-Agile team by creating what the User Stories requested. Without the role of a developer, there would have been no software to deliver to the stakeholders. Developers also contributed by communicating with the rest of the team and estimating the timeframe needed to complete each User Story (Grebić, 2021). In our project, the developers communicated their progress during the Daily Scrum meetings by explaining what they had completed, what they were currently working on, and whether they had encountered any problems. Another example was when the developer created the slideshow that was requested by the end user. Overall, the developer's role was the backbone of the project and was essential for delivering a completed product.

The tester was another role that contributed to the Scrum-Agile team. Testers specifically helped the developers by creating test cases, guiding the development team toward meeting the required criteria, and ensuring that the project functioned correctly. Testers also communicated closely with developers. If there was a change in the priorities of the project, they would adjust their test cases and testing criteria accordingly (Grebić, 2021). For example, when we initially created the test cases, they were designed for a website format. However, the Product Owner later informed us that we would be switching to a slideshow format. As a result, the tester had to modify the test cases to better fit the new criteria. This allowed the tester to verify that the slideshow met the updated requirements instead of testing functionality that was no longer part of the project. Overall, the tester contributed to the team by helping developers meet the project's requirements and quality standards, resulting in more reliable and coherent software.

The next role, the Scrum Master, also contributed significantly to the Scrum-Agile team. The main responsibility of the Scrum Master was to provide support to the rest of the team by helping bridge the gap between the Product Owner and the development and testing teams. In our project, the Scrum Master gathered the team each day for a Daily Scrum meeting that lasted approximately fifteen minutes. During these meetings, the team discussed three main questions: What did you do yesterday? What will you do today? What impediments are preventing you from making progress? These meetings allowed our team to communicate progress and identify problems that could prevent User Stories from being completed. At the end of a sprint, the Scrum Master conducted the Sprint Review and Retrospective to inspect what was completed and determine whether anything could be improved for future sprints. These meetings also provided an opportunity to communicate changes to the team based on stakeholder feedback (Grebić, 2021). Overall, the Scrum Master's main contribution to our project was serving as a connection between the development team, Product Owner, and stakeholders.

The last role of the Scrum-Agile team was the Product Owner. The main contribution of the Product Owner was maintaining constant communication between the stakeholders and the Scrum Master and development team. As stated by Magnussen et al. (2024), the Product Owner is a key communicator and prioritizer who is responsible for improving collaboration between stakeholders and the Scrum team. The Product Owner helps ensure that requirements are understood correctly and that the development team creates a product that aligns with the stakeholders' expectations. A direct example of this in our project occurred when the Product Owner attended a meeting with the stakeholders and listened to their ideas regarding the direction of the project. Later, the Product Owner communicated this information to the Scrum Master, who then directed the rest of the team toward meeting the stakeholders' vision. Another

important task performed by the Product Owner was managing and grooming the backlog. Specifically, the Product Owner prioritized the User Stories based on what the stakeholders wanted during the most recent meeting. This helped our team focus on the features that were most important to the stakeholders instead of spending time on lower-priority features.

Completing User Stories

The completion of User Stories was more effective when using the Scrum-Agile methodology. It allowed User Stories to be fully completed rather than leaving a large number of partially completed stories. This approach encouraged our team to focus on completing work before moving on to additional User Stories. The main factors influencing completion were the ranking and prioritization of the backlog and the guidance that the Scrum Master provided to the team during the Daily Scrum meetings. Because User Stories were prioritized in the backlog, the development team had a clear understanding of which features needed to be completed first.

If we consider traditional methods such as the Waterfall methodology, we will spend much of our time planning the implementation of the User Story before developing the product. However, since we used the Agile methodology, we spent more time working toward a shippable product rather than focusing primarily on planning. Therefore, at any point during the development process, we could have a product demonstrated to the stakeholders. Agile also allowed us to respond to changes from the stakeholders without having to completely restart the project. This helped our team continue completing User Stories while adapting the project to new requirements.

In our project, one of the stakeholders asked whether we could include links or a list that ranked the best vacation destinations to visit. The Product Owner created a User Story for the

development team based on this request. The User Story was then added to the backlog and prioritized according to the stakeholders' needs. Because the User Story was clearly defined and prioritized, the development team had a better understanding of what needed to be implemented. In our case, the first implementation involved creating a ranking of the best vacation destinations, along with another User Story that provided a link to additional information about each destination. After the development team implemented the features, the tester could verify that they met the requirements. This process allowed the User Story to move from a stakeholder request to a defined requirement, then to development and testing, and ultimately toward completion. This demonstrates how the Scrum-Agile approach helped our team consistently turn stakeholder requests into completed features.

Handling Interruptions

For a typical Scrum-Agile team, they usually run into interruptions such as the stakeholders wanting to change the direction of the project. Typically, it is handled by having a high number of meetings, including the daily standups, planning sessions, and the burndown or burnup charts (Wiesche 2021). Those characteristics are an important and essential role for the development team to change direction and include what the stakeholders want at any given time.

In my scenario, the stakeholders wanted to change a large aspect of the project mid-development. Instead of using a website approach for the SNHU travel company, they wanted to use a slide show to show off the top 5 best destinations to visit. While still including the main core concepts such as a link, image, and a short description of the destination.

This change was conducted because of the constant communication between the Product Owner, Scrum Master, and the development/testers. We held constant meetings, which allowed the rest

of the team to understand the new direction the project is going. Also, the Product Owner, development team, and the testers constantly emailed each other any questions regarding the new changes. Overall, the interruption of the project was a minor setback, as we managed to use some of the previous code since it was modularly developed.

Communication

Communication is an essential component of Scrum-Agile development because it allows team members and stakeholders to remain aligned throughout the development process. During the SNHU Travel project, our team primarily communicated through Daily Scrum meetings and email communication. These methods allowed team members to share progress, discuss challenges, and request clarification when requirements were unclear.

One example of effective communication occurred when the tester reached out to the Product Owner for additional clarification regarding user stories and testing requirements. This communication was effective because it asked specific questions related to unclear requirements instead of making assumptions. By identifying areas of uncertainty early, the team was able to better understand stakeholder expectations and create more accurate test cases.

This example also supported important Scrum principles such as transparency, collaboration, and adaptability. Transparency was demonstrated because the tester openly communicated areas where requirements were unclear. Collaboration was encouraged because the tester, Product Owner, and development team worked together to improve understanding of the requirements. Additionally, this communication allowed the team to adapt before development was completed, reducing the possibility of needing major changes later in the project.

The following email demonstrates this communication:

Example email:

Subject: Clarification Needed for User Stories and Testing Requirements

Hello Product Owner,

I have a few questions regarding the current user stories that would help guide the development of my test cases. After reviewing the requirements, I noticed there are a few areas where additional clarification would help ensure the final product meets stakeholder expectations.

First, I noticed that the priority of each user story was not included. Could you provide the ranking or priority level for each story? This information would help me determine which requirements should receive the most testing attention.

For User Story #1:

Could you clarify the expected image size and formatting requirements? During testing, I noticed that image dimensions could vary depending on the implementation. Having a standard size or guideline would help ensure consistency across the application. Additionally, are there specific expectations for text placement and font size?

For User Story #2:

The user story includes a list of required attributes, but it does not specify where those attributes should appear on the page. The development team placed the attribute section in the center of the screen. Could you confirm whether this matches stakeholder expectations or if there is another preferred location?

For User Story #3:

I noticed that this feature uses a slide-based format. Should there be a main navigation page that introduces the available options, with each slide representing a different ranking category based on price? Or should users immediately enter the ranking system?

Having clarification on these items will allow me to create more accurate test cases and ensure the product aligns with stakeholder expectations.

Thank you for your assistance.

Josh Velazquez

The email was effective because it provided detailed questions that the Product Owner could easily respond to. Instead of asking broad questions, the tester identified specific concerns for each user story, including image formatting, attribute placement, and slide navigation requirements. This made it easier for the Product Owner to provide useful feedback and helped ensure that the final product would align with stakeholder expectations.

Overall, this communication approach improved collaboration among team members by encouraging open discussion and reducing uncertainty. In a Scrum-Agile environment, frequent and clear communication is necessary because it allows teams to continuously improve, address issues quickly, and deliver a product that better meets stakeholder needs.

Organizational Tools

A common Scrum communication practice that creates openness and transparency within a team is the Sprint Review. This event allows the team and stakeholders to evaluate completed work

from the previous iteration and determine whether the deliverables meet expectations. During the Sprint Review, team members can discuss completed user stories, identify areas for improvement, and gather feedback that can influence future iterations. This process supports the Scrum principle of continuous improvement because the team is constantly evaluating the effectiveness of their work.

According to the study of *The Effects on Social Support and Work Engagement with Scrum Events*, Scrum events can improve collaboration and strengthen team relationships by creating supportive working conditions. This demonstrates how frequent communication and structured Scrum practices help teams address problems earlier. When team members have transparency and psychological safety, they are more likely to communicate challenges with their assigned tasks rather than attempting to solve problems individually. This allows the Scrum team to work together to remove impediments and improve the development process (Muller, 2021).

The organizational tools that contributed most to the success of the SNHU Travel project were the Agile Team Charter, Daily Scrum Meeting practices, Sprint Planning activities, and Jira. The Agile Team Charter helped establish project goals, define team expectations, identify potential risks, and ensure that everyone understood their responsibilities. This supported Sprint Planning because the team had a clear understanding of the project's objectives before assigning tasks.

The Daily Scrum Meeting was another important Scrum event because it allowed team members to communicate progress, discuss completed work, and identify any obstacles preventing tasks from being completed. If a team member encountered an impediment, the Daily Scrum provided a structured opportunity to bring attention to the issue so the team could collaborate on a solution.

Jira was an effective Agile project management tool because it allowed the team to organize user stories, assign tasks, track progress, and manage issues throughout development. Jira supported multiple Scrum events by helping the team prepare during Sprint Planning, monitor progress during the Sprint, and review completed work during Sprint Reviews. Jira was especially useful during Sprint Planning because it allowed the team to organize and prioritize user stories, while the Daily Scrum helped us monitor the progress of those tasks throughout the sprint. By having a visual representation of tasks and their current status, team members could easily understand what work needed to be completed without relying on constant communication with other team members (Zelenika, 2026).

Overall, the combination of Scrum principles and organizational tools improved the team's ability to communicate, manage responsibilities, and complete project tasks. The structured nature of Scrum events provided the team with consistent opportunities to plan, review progress, and make improvements throughout the SNHU Travel project.

Evaluating Agile Process

The Scrum-Agile process used during the SNHU Travel project presented several strengths as well as some weaknesses. Like any software development methodology, Scrum-Agile has advantages and disadvantages depending on the needs and complexity of the project. The most important factor is selecting an approach that best matches the project's requirements.

One of the major advantages of using Scrum-Agile for the SNHU Travel project was its ability to adapt to changes. The team could adjust priorities, modify features, and respond to stakeholder feedback throughout development. Scrum also encouraged modular development, which allowed different sections of the project to be worked independently. Another benefit was the

communication between the development team and stakeholders, which helped ensure that the final product aligned with expectations. Additionally, having working features available to demonstrate throughout development provided opportunities for feedback before the project was completed.

However, the Scrum-Agile approach also presented some challenges. Implementing Scrum can be difficult because it requires strong communication, organization, and commitment from every team member. Frequent changes from stakeholders can also increase development costs and extend the project timeline. Additionally, Agile depends heavily on collaboration, meaning that if one part of the team does not complete their responsibilities, it can negatively impact the entire project.

For the SNHU Travel development project, I believe a traditional Waterfall approach may have been a better fit. The project had a relatively simple scope, with a clear goal of displaying travel destinations and ranking recommended locations. Since there was limited uncertainty and not many complex features, more upfront planning could have helped define requirements before development began.

Although Scrum-Agile was effective for managing communication and completing incremental features, the project did not require frequent changes or major pivots. For example, when the project transitioned to the slideshow feature, those requirements could have been identified earlier through additional planning and stakeholder discussions.

Overall, Scrum-Agile would be more beneficial for a larger-scale project involving many features, user stories, and changing requirements. While Agile provided useful collaboration and

development practices for the SNHU Travel project, a Waterfall approach may have been more efficient because the project's requirements were more predictable and easier to define upfront.

References

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